SHREE SANJEY S K

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SUMMARY

- Software engineer focused on the systems behind AI products: real-time video analytics, distributed inference pipelines, and RAG applications. Currently on my internship semester at Katomaran Technologies, building intrusion, crowd, and perimeter-crossing analytics on live camera feeds. M.Sc. Software Systems student at PSG Tech (graduating 2028).

SKILLS

Languages: Python, Java, C++, SQL, JavaScript, TypeScript

AI / ML: Scikit-learn, PyTorch, NLTK, SHAP, OpenCV, YOLO, vLLM

Frameworks: Next.js, React, FastAPI, Django, Flask, Tailwind CSS

Databases: PostgreSQL, MongoDB, SQLite, Redis

Tools and Infrastructure: Docker, Git, Vercel, RabbitMQ, AWS S3

EXPERIENCE

Katomaran Technologies – AI & Video Analytics Intern *Coimbatore, India | May 2026 – Present*

- Independently developed 5 real-time video analytics for the Odisha Prison Project: intrusion, wall climbing, crowd detection, perimeter crossing, and object-thrown detection.
- Contributed to redesigning a distributed video processing pipeline with motion-triggered recording and YOLO-based inference workers for person and vehicle detection, and assisted in model training and dataset preparation for GD Naidu ITMS.
- Evaluated video-language models using vLLM, developed AI-powered incident search, and contributed to an Orange Pi edge gateway for camera streaming, device management, and cloud onboarding.

PROJECTS

Placement Chronicles – PSG Placement Experience Platform [Live] *Next.js, TypeScript, MongoDB, Gemini*

- Built a live platform used by 400+ PSG Tech students, with 100+ interview experiences uploaded. Features AI-powered PDF ingestion (Gemini 1.5 Flash) to extract structured interview data, a RAG-powered chatbot for semantic search across all experiences, vector embeddings with MongoDB Atlas Vector Search, Upstash Redis rate limiting, and a 3D starfield UI with Framer Motion animations.

PlanB – Autonomous AI Scheduling Assistant *Python, FastAPI, LangGraph*

- Built an autonomous scheduling assistant operating entirely through WhatsApp — no app to open, no manual rescheduling. Orchestrates 14 specialized LangGraph agents (Groq llama-3.3-70b-versatile + llama-3.1-8b-instant, Gemini 1.5 Flash) through a dynamic state machine for disruption detection, cascade impact mapping across all calendar events, and calendar rescheduling via Google Calendar API. Audit logs persisted to AWS S3.

FaceTracker – Real-Time AI Face Detection & Visitor Counting *Python, FastAPI, React, PostgreSQL*

- Built a real-time face detection, recognition, and visitor counting system for the Katomaran Hackathon using YOLOv11, ByteTrack, InsightFace, and FAISS. Supports up to 4 simultaneous RTSP/video streams with automatic registration of new visitors, WebSocket push of live entry/exit events (zero polling), 18 REST endpoints, and a React + Recharts monitoring dashboard.

EDUCATION

College **Present**

PSG College of Technology – M.Sc (Integrated) Software Systems *CGPA: 6.62 / 10*

High School **May 2023**

Sri Kumarans Composite PU College, Bangalore *Percentage: 89.16*

Secondary School **May 2021**

Sri Kumarans Children's Academy, Bangalore *Percentage: 88.00*